The Campus of Tomorrow

Higher Colleges of Technology

كليات التقنية العليا

CIS 2033-USER CENTERED DESIGN

Week 5 User Research

Week 5	**CLO 3: Apply user research techniques to understand user needs and develop user personas**

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	28/05/2025	Define the first version	001

X.Y
X: Major change
Y: minor change

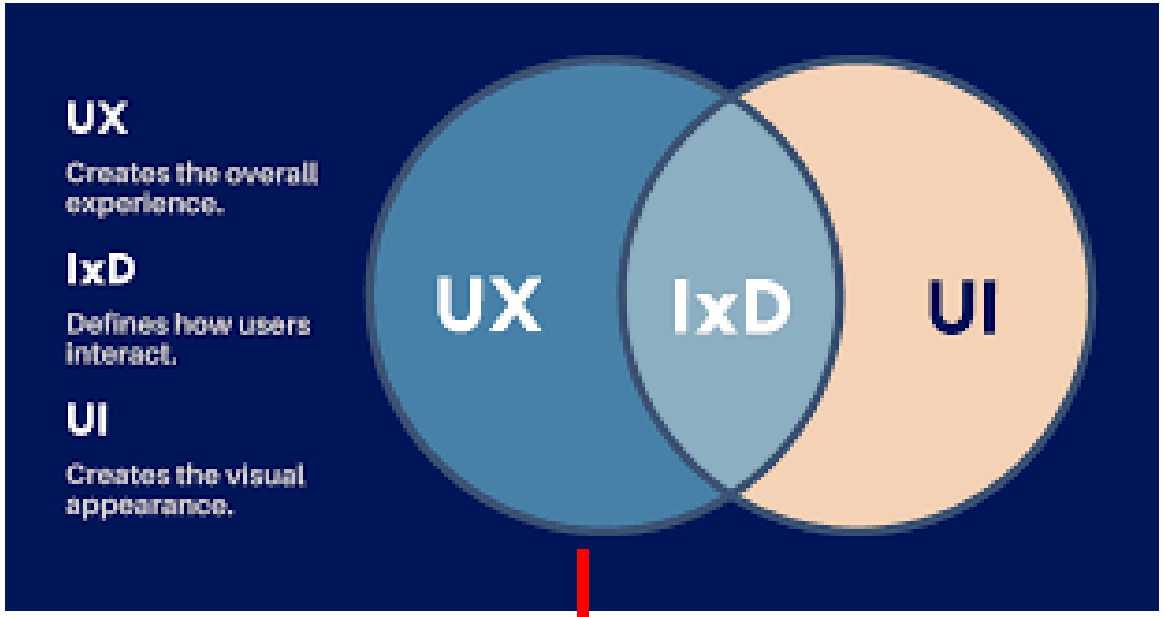
Recap

- So far, we
  - Understood the User-Centered Design (UCD) process
  - Learned how to apply ethical practices and sustainability in UCD
  - Explored the concept of accessibility for people with disabilities (PwD)
  - Understood how heuristic evaluation is used to assess usability

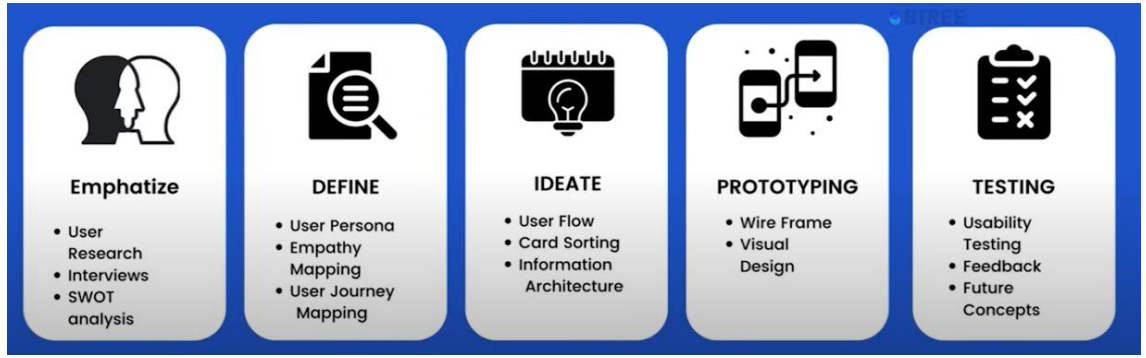
Reflection Questions

- Do you know how to start the UCD process with the empathise phase?
- Can you explain when, where, and why to conduct user research?
- Do you understand how to analyze your users or stakeholders?
- Do you know how to create a user recruitment plan for your research?

How far have we come?



We are here now



We are here now

Lecture Outline

- Understanding Requirements
- Types of Requirements
- Introduction to user Research
- Stakeholder Analysis
- Recruiting users
- User Research Methods
- Ethical Research Practices

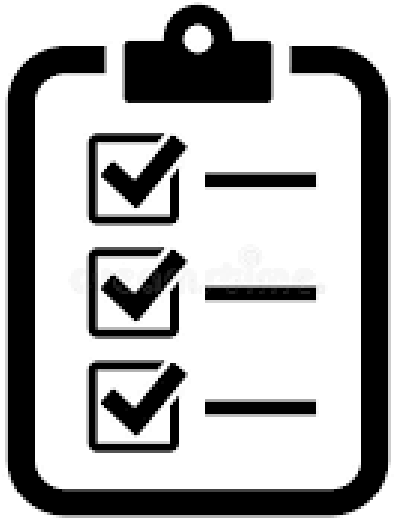
LECTURE OBJECTIVES

- At the conclusion of the lecture, the student should be able to perform the tasks below:
  - Understand the significance of user requirements
  - Identify types of functional and non-functional requirements
  - Analyze stakeholder types and create a stakeholder map
  - Apply user research methods: interviews, observations, contextual inquiry
  - Evaluate and ensure ethical research practices
  - Create thematic insights using qualitative data analysis

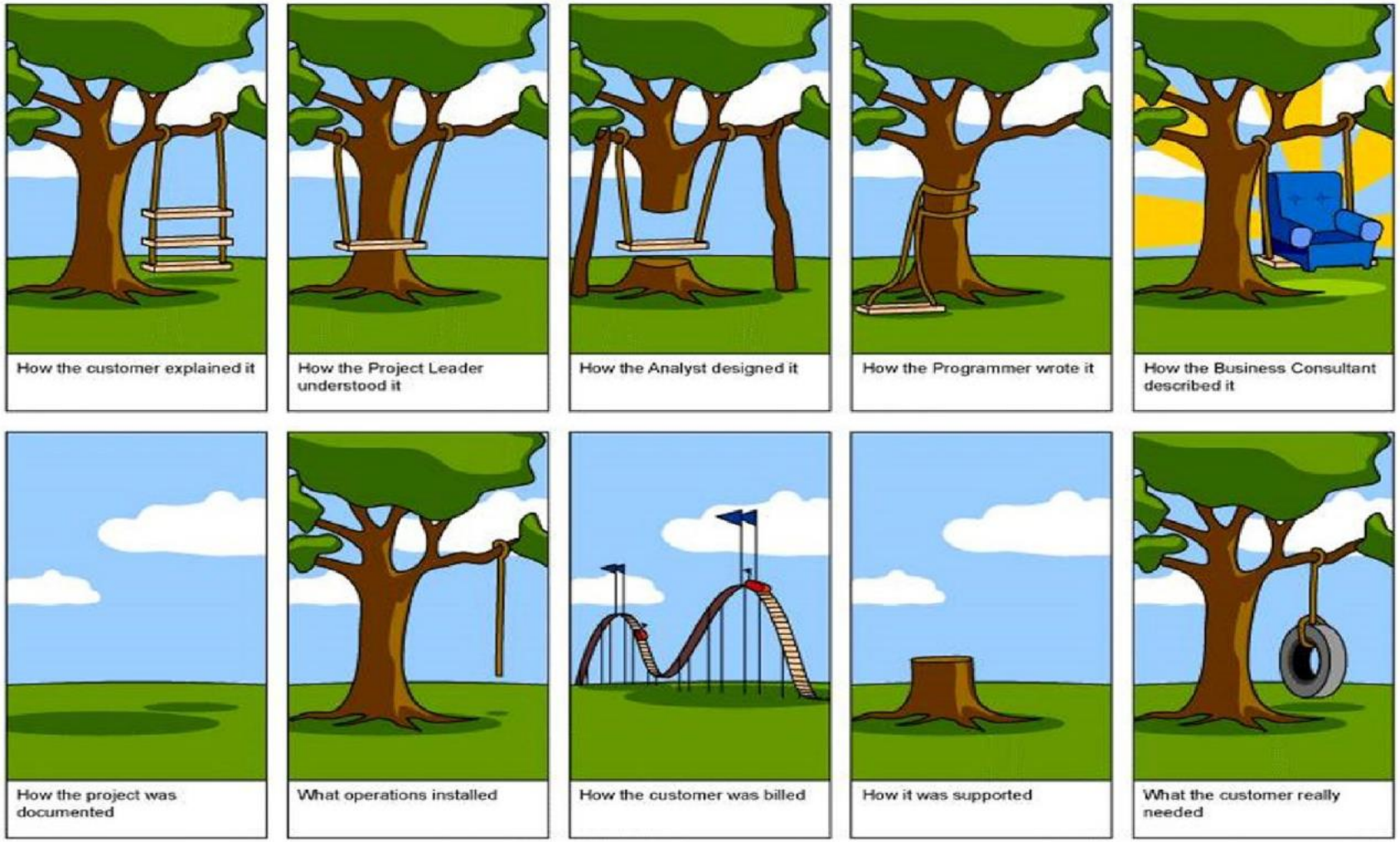
Requirement Gathering

What is a Requirement?

- A REQUIREMENT is something a product must do or a quality it must have.
- Key Questions:
  - What should the software do?
  - How will it meet the user's needs?
  - What makes it a good product (fast, easy to use)?
  - Are there limits or rules?



What is a Requirement?



What is a Requirement?

- Things can go wrong:
  - Customers may not explain needs clearly.
  - Designers might misunderstand.
  - Programmers could build the wrong thing.
- Requirements analysis helps:
  - Clearly explain what the system will do.
  - Show how it will work for users.
  - Do this before building anything!

Types of Requirements

- Functional: What the system should do (e.g., "Users can create an account").
- Non-Functional: How the system behaves (e.g., "Website loads in 2 seconds").
  - Data: What kind of information is needed.
  - Environmental: Conditions in which the system operates.
  - User Characteristics: Who will use it.
  - Usability & UX Goals: Ease and joy of use.

⚠️ **Don't assume functional is more important than non-functional. Both matter!**

Types of Requirements

- If you are developing a new product, you should define the functional requirements yourself.
- If you are enhancing an existing product, refer to the current functional requirements provided.
- As part of your UX design process, conduct user research (CLO3), identify user needs, and use those insights to generate and update the functional requirements accordingly.
- During Weeks 5 and 6, we will conduct user research.
- In Weeks 7 and 8, we will develop user personas, empathy maps, and user journey maps to identify user requirements.
- Based on these insights, we will then revise and update the existing functional requirements for UX design.

Functional Requirements of Skyline Airlines

- Users can search for flights by date, origin, destination, and fare type.
- Flight details like schedule, layovers, aircraft, and seats are shown.
- Users can pick seats and choose upgrades or extras.
- Bookings can be made, changed, or canceled.
- Payments are handled securely online.
- Users can create accounts, save details, and view past bookings.
- Support is available through FAQs, contact forms, or chat.
- Users can request travel assistance if needed.
- Seat availability is updated in real time.
- The system provides reports on bookings and travel trends.

⚠️ **Since we are redesigning the existing Skyline Airlines website with a focus on UX, the provided functional requirements reflect those in the current site and will update after conducting use research process from wk5-8.**

Non-Functional Requirements

- Data Requirement
  - Where does the DATA come from?
  - What needs to be saved?
  - How long will it be stored?
  - How correct must it be?
  - How will it be shown?

- Environmental
  - Physical: Is it cold, crowded, noisy?
  - Social: Teamwork needed?
  - Organizational: Single user or company-wide?
  - Technical: Must work with other systems?

Non-Functional Requirements

- User Characteristics: Things about the user that affect product design:
  - Tech Skills – Are they good with computers?
  - Personal Info – Age, education, location.
  - Abilities – Any physical or mental disabilities.
  - Preferences – Likes, dislikes.

Non-Functional Requirements

- Usability Goals: How easy and good is the product to use?
  - Effective – Does it do the job?
  - Efficient – Is it quick and easy?
  - Safe – Can users avoid errors or harm?
  - Useful – Does it offer good features?
  - Easy to Learn – Can beginners pick it up fast?
  - Easy to Remember – Can users recall how to use it after time away?
- User Experience Goals: How does the product make users feel?
  - Enjoyable
  - Entertaining
  - Visually Appealing
  - Motivating

Lab Activity : Functional & Non-Functional Requirements

Take a quick look at the lab activity on functional and non-functional requirements for Skyline Airlines.

Then, discuss with your project group how you plan to apply these concepts to your own project.

<<see next slide for the project activity>>

Project Activity: Exploring Functional and Non-Functional Requirements

1. Explore: List all the functional requirements of your system

2. Explore: What are the non-functional requirements of your system based on the following dimensions:
a. What data is required and how will it be handled?
b. What user characteristics are most relevant to your project?
c. What usability goals do you aim to achieve with your design?

Check the Lab Activity done for skyline airlines(CLO3_1 Functional and Non-Functional requirement -Skyline Airlines)

Instructions:

- Discuss each question with your group
- Write down examples specific to your case study
- Be ready to share your findings with the class!

Key Points to Remember

- We haven't conducted user research yet to fully understand the user requirements.
- You initially listed functional and non-functional requirements.
- These requirements should be updated regularly during each stage of the UCD process.
- For example, while interacting with Skyline users, you might discover new needs.
- During testing, if users struggle with certain tasks, that feedback should update the requirements list.

User Research

What is User Research?

- User research is the process of learning about your users –
  - who they are,
  - what they do,
  - what they need, and
  - what problems they face.
- We use different methods like
  - interviews, surveys, and testing to gather this information.

Why User Research Matters?

- Know what users need and like
- Design products that are easy and fun
- Show return on investment (ROI) through good design

ROI (Return on Investment) in software development measures how much profit or value a software project generates compared to its cost.

When to Do User Research?

- USER RESEARCH should happen at every stage:
  - Before Designing – Learn what users need
  - During Development – Test and improve
  - After Release – See how it's used, improve again

Lab Activity : User Research

Take a quick look at the lab activity on User research for Skyline Airlines.

Then, discuss with your project group how you plan to apply these concepts to your own project.

<<see next slide for the project activity>>

Project Activity: Importance of User Research

- **Engage**: Why is user research important in the context of your selected case study? Provide an example of how it will impact your design.

- **Explain**: At what stage of the Software Development Life Cycle (SDLC) will you carry out your user research? Provide reasoning for your choice and how it supports the design process.

Check the Lab Activity done for skyline airlines – CLO3_2 User Research for Skyline Airlines

Instructions:

- Discuss each question with your group
- Write down examples specific to your case study
- Be ready to share your findings with the class!

Stakeholder Analysis

Stakeholder Analysis: Types of Stakeholders(1)

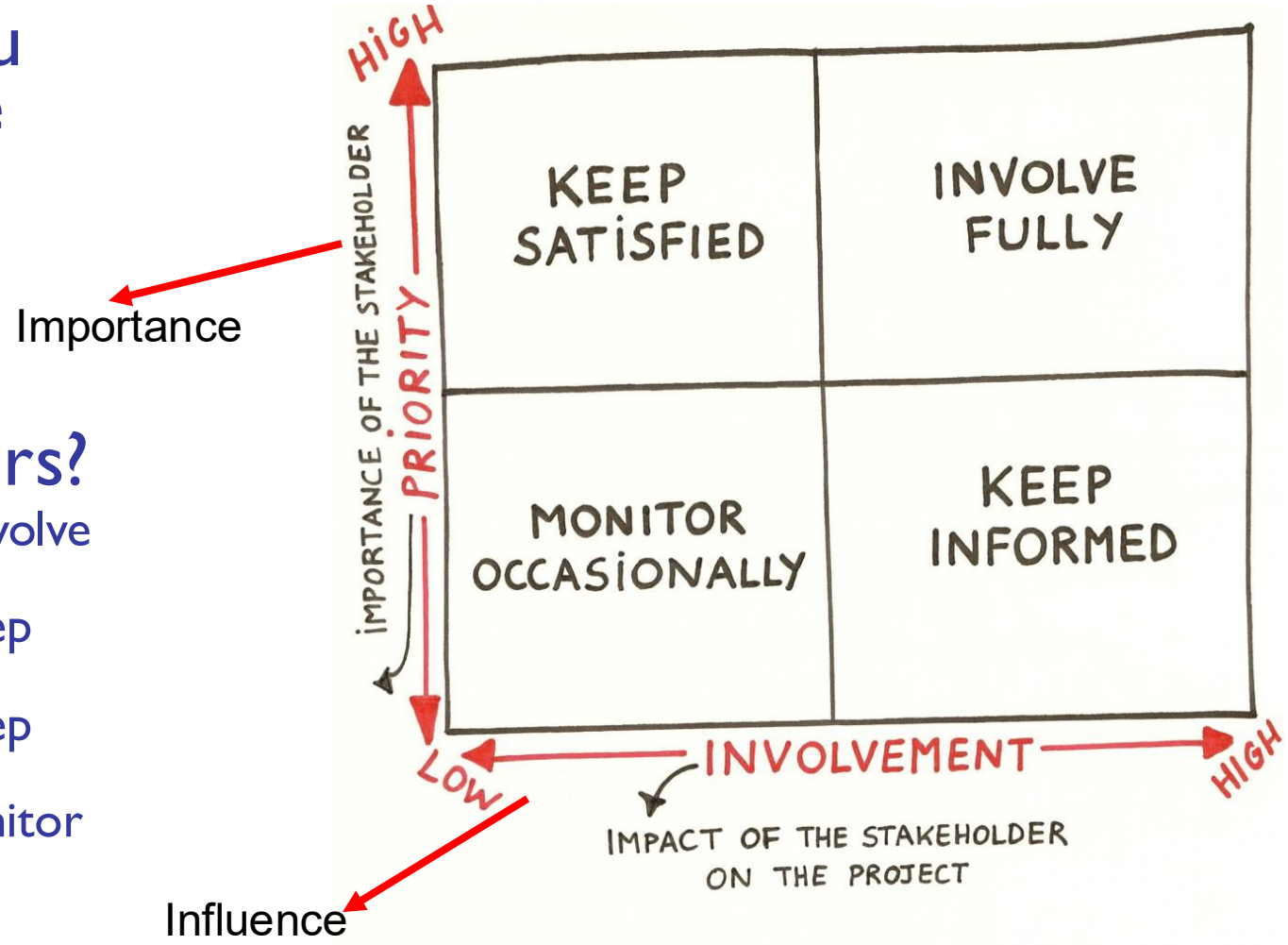
- Who is affected by your system?
  - Primary: Main users (e.g., customers)
  - Secondary: Support Indirectly affected users (e.g., staff)
  - Tertiary: (e.g., managers)
  - Facilitators: Builders (e.g., developers, IT)

Stakeholder Analysis: Types of Stakeholders (2)

- Another way to find stakeholders is by looking at the organization itself.
  - Internal Stakeholders: Employees, Managers, Owners, Contractors
  - External Stakeholders: Customers, Society, Suppliers, Investors, Government

Stakeholder Analysis: Stakeholder Mapping

- Stakeholder mapping helps you figure out how much influence different people have on your project. It also shows how important they are to your organization.
- How to handle the stakeholders?
  - **High Importance / High Influence:** Involve fully
  - **High Importance / Low Influence**: Keep satisfied
  - **Low Importance / High Influence**: Keep informed
  - **Low Importance / Low Influence**: Monitor occasionally



Stakeholder Analysis

- Tips for identifying participants:
  - Focus on target users
  - Include diverse groups
  - Avoid bias (don't use insiders or friends)

Lab Activity : Stakeholder Analysis

Take a quick look at the lab activity on stakeholder analysis for Skyline Airlines.

Then, discuss with your project group how you plan to apply these concepts to your own project.

<<see next slide for the project activity>>

Project Activity: Stakeholder Analysis

- **Explore**: Identify the stakeholders for your project and classify them into the following categories:
  a. Primary, Secondary, and Tertiary
  b. Internal and External stakeholders

- **Elaborate**: After identifying your stakeholders, develop a stakeholder map showing how you will prioritize them based on their influence and interest in the project.

Check the Lab Activity done for skyline airlines - CLO3_3 Stakeholder Analysis for Skyline Airlines

Instructions:

- Discuss each question with your group
- Write down examples specific to your case study
- Be ready to share your findings with the class!

Recruiting the User

What is User Research Recruitment?

- Finding people who are like our real users to participate in research studies.
- These participants should have the same qualities and experiences as our target customers.

Why User Recruitment is Important?

- Accurate insights = better decisions
- Understand real user needs
- Build products that fit their lives

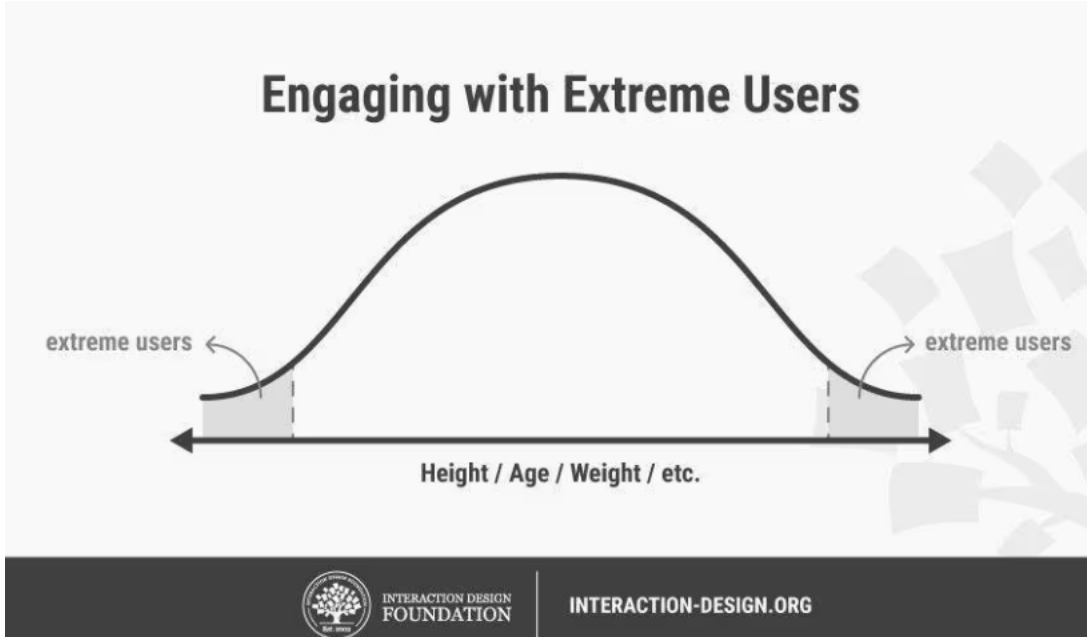
Who Should You Recruit?

- Real Users – actual or typical users
- Diverse Groups – reflect different needs
- Avoid bias – no insiders or overly informed people

Who Should You Recruit?

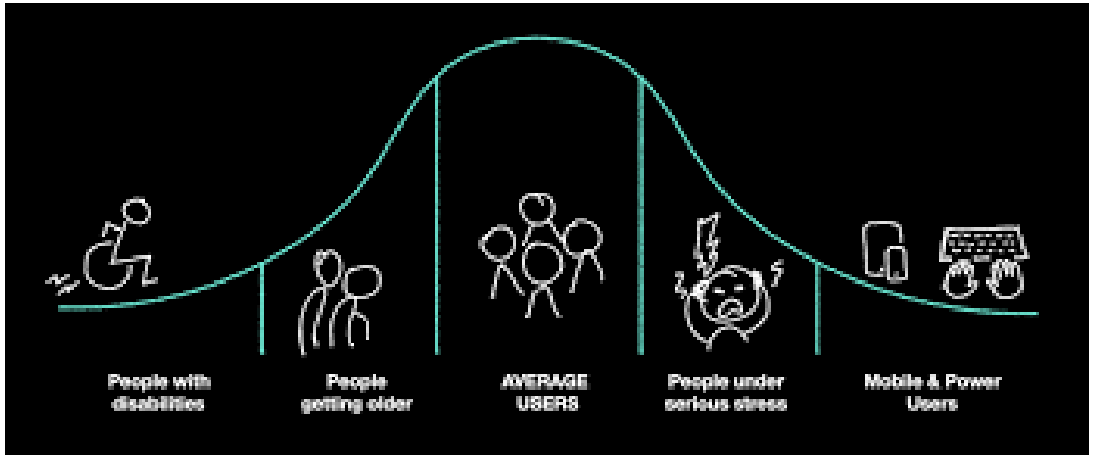
Why Engage Extreme Users?

- **Extreme users** are those who differ significantly from average users (e.g., shortest/tallest, tech-savvy/newbies). They help uncover **edge-case problems** that average users might not reveal.
- Designing for **extreme users** improves usability for everyone. If a design works well for them, it's usually more **inclusive and user-friendly** for all.



Group Activity: Who Should You Recruit?

- Can you identify who the extreme users might be for the Skyline project?
- Are you able to determine them for your own project as well?



Tips for Successful Recruitment

- Set clear, balanced criteria
- Recruit extra people (in case of no-shows)
- Pay participants
- Test your screener for clarity

Lab Activity : User Recruitment Plan

Take a quick look at the lab activity on user requirement plan for Skyline Airlines.

Then, discuss with your project group how you plan to apply these concepts to your own project.

<<see next slide for the project activity>>

Project Activity: User Recruitment Plan

- **Explore**: What type of users will you recruit for your user research? Justify their relevance to your system's objectives and design scope.

- **Elaborate**: Describe the process and steps you will follow to ensure successful and ethical user recruitment. Include any tools or techniques you will use.

Check the Lab Activity done for skyline airlines - User Recruitment for Skyline Airlines(CLO3_4)

Instructions:

- Discuss each question with your group
- Write down examples specific to your case study
- Be ready to share your findings with the class!

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Thank You